

## Network Access Device:

- o NAD is a network term, which stands for **Network Access Device**.
- o Network Access Device (NAD) is the OSI Model Access Layer Device.
- o NAD is basically, a device through which the Endpoint is connected.
- o Network Access Device (NAD) is AAA Client device where endpoint connected.
- o Any device that is going to send RADIUS authentication requests to Cisco ISE.
- o NAD responsible for sending EAP authentication request to authentication server.
- o NAD is Authenticator for 802.1X from a supplicant to the authentication server.
- o NAD can be the device such as Switch, Router that support NAC Enforcement policies.
- o Common NAD types Wired Ethernet Switches, Wireless LAN Controllers & ASA.



Wireless LAN Controller

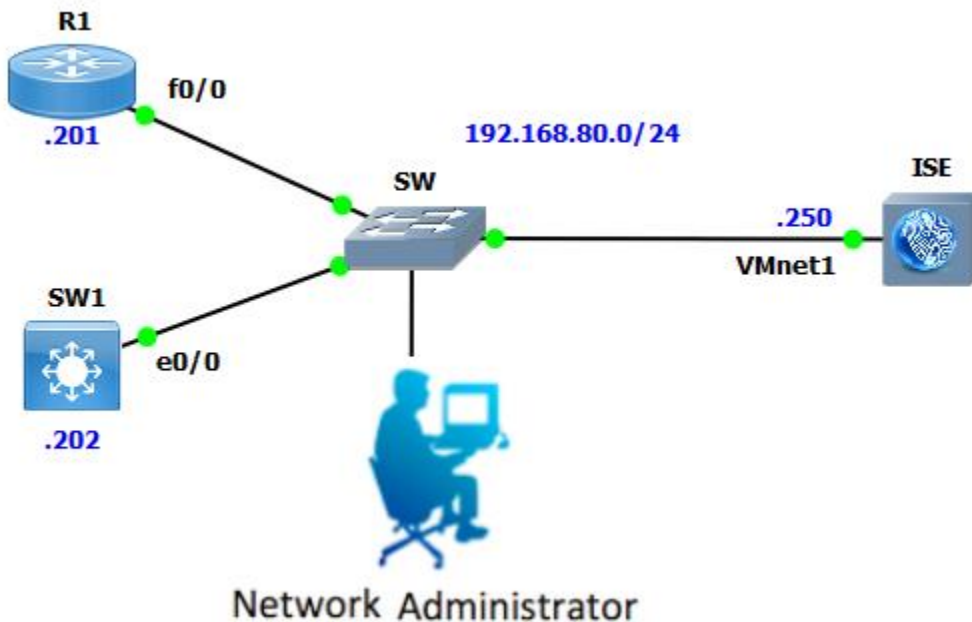
Switch

Router

The screenshot shows the Cisco Identity Services Engine (ISE) Administration interface. The navigation pane on the left has 'Administration' expanded, with 'Network Resources' selected. Under 'Network Resources', 'Network Devices' is highlighted. The main content area shows the 'Network Devices' page with a table of existing devices. A red box highlights the 'Add' button in the toolbar, and a green box highlights a row in the table representing a device named 'SW1' with IP '192.168.80.1...' and profile 'Cisco'.

| Name | IP/Mask         | Profile Name | Loca  |
|------|-----------------|--------------|-------|
| SW1  | 192.168.80.1... | Cisco        | All L |

## Network Diagram:



## Configure Router for Authentication and Authorization:

Enable AAA new-model. Define TACACS server ISE. Configure login, enable authentications, and then use the exec and command authorizations. Assign method lists to line vty.

|                                                                      |
|----------------------------------------------------------------------|
| <b>R1 AAA Configuration</b>                                          |
| R1(config)#aaa new-model                                             |
| R1(config)#tacacs-server host 192.168.80.250 key test123             |
| R1(config)#aaa authentication login default group tacacs+ local      |
| R1(config)#aaa authentication enable default group tacacs+ enable    |
| R1(config)#aaa authorization exec default group tacacs+ local        |
| R1(config)#aaa authorization commands 0 default group tacacs+ local  |
| R1(config)#aaa authorization commands 1 default group tacacs+ local  |
| R1(config)#aaa authorization commands 15 default group tacacs+ local |
| R1(config)#aaa authorization config-commands                         |
| R1(config)#line vty 0 4                                              |
| R1(config-line)#authorization commands 0 default                     |
| R1(config-line)#authorization commands 1 default                     |
| R1(config-line)#authorization commands 15 default                    |
| R1(config-line)#authorization exec default                           |
| R1(config-line)#login authentication default                         |

### Configure Switch for Authentication and Authorization:

Enable AAA new-model. Define TACACS server ISE. Configure login, enable authentications, and then use the exec and command authorizations. Assign method lists to line vty.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SW1 AAA Configuration</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| SW1(config)#aaa new-model                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| SW1(config)#tacacs server ISE<br>SW1(config-server-tacacs)#address ipv4 192.168.80.250<br>SW1(config-server-tacacs)#key test123                                                                                                                                                                                                                                                                                                                                                    |
| SW1(config)#aaa authentication login default group tacacs+ local<br>SW1(config)#aaa authentication enable default group tacacs+ enable<br>SW1(config)#aaa authorization exec default group tacacs+ local<br>SW1(config)#aaa authorization commands 0 default group tacacs+ local<br>SW1(config)#aaa authorization commands 1 default group tacacs+ local<br>SW1(config)#aaa authorization commands 15 default group tacacs+ local<br>SW1(config)#aaa authorization config-commands |
| SW1(config)#line vty 0 4<br>SW1(config-line)#authorization commands 0 default<br>SW1(config-line)#authorization commands 1 default<br>SW1(config-line)#authorization commands 15 default<br>SW1(config-line)#authorization exec default<br>SW1(config-line)#login authentication default                                                                                                                                                                                           |

### Add Local Groups & Users:

Navigate to **Administration > Identity Management > Groups > User Identity Groups**. Click **Add**. Provide Name, Description and click **“Save”** to save setting.

The screenshot displays the Cisco Identity Services Engine (ISE) Administration interface. The top navigation bar includes 'Home', 'Operations', 'Policy', 'Guest Access', and 'Administration'. The 'Administration' menu is expanded, showing 'System', 'Identity Management', 'Network Resources', 'Device Portal Management', 'pxGrid Services', and 'Feed S'. The 'Identity Management' menu is further expanded, showing 'Identities', 'Groups', 'External Identity Sources', 'Identity Source Sequences', and 'Settings'. The 'Groups' menu is selected, and the 'User Identity Groups' sub-menu is highlighted. The main content area shows the 'User Identity Groups > AAA-Group' configuration page. The 'Identity Group' section has a '\* Name' field with the value 'Admin-Group' and a 'Description' field with the value 'This is Admin Group'. The 'Save' button is highlighted, and the 'Reset' button is also visible. Below the configuration fields, the 'Member Users' section is partially visible.

Navigate to **Administration > Identity Management > Groups > User Identity Groups**. Click **Add**. Provide Name, Description and click **“Save”** to save setting.

The screenshot shows the Cisco ISE Administration console. The breadcrumb navigation at the top is: **Administration > Identity Management > Groups > User Identity Groups**. The left sidebar shows the navigation tree with **Groups** selected under **Identity Management**. The main content area is titled **User Identity Groups > New User Identity Group**. It contains the following fields and actions:

- Identity Group**: Labeled with a red '5', this section contains the **\* Name** field with the value 'Support-Group' and the **Description** field with the value 'This is Support Group only Show Commands'.
- Submit** and **Cancel** buttons are at the bottom of the form.
- Red numbers 1 through 7 highlight the navigation steps and form fields.

Navigate to **Administration > Identity Management > Identities > Users**. Click **Add**. Provide Name, Passwords and User Groups select Admin-Group and click **“Save”** to save the setting.

The screenshot shows the Cisco ISE Administration console. The breadcrumb navigation at the top is: **Administration > Identity Management > Identities > Users**. The left sidebar shows the navigation tree with **Users** selected under **Identities**. The main content area is titled **Network Access Users List > admin1**. It contains the following sections and fields:

- Network Access User**: Labeled with a red '5', this section contains the **\* Name** field with the value 'admin1' and the **Status** dropdown set to 'Enabled'.
- Passwords**: Labeled with a red '6', this section contains the **\* Login Password** and **Re-Enter Password** fields, both masked with dots.
- User Information** and **Account Options** sections are visible but empty.
- User Groups**: Labeled with a red '7', this section contains a dropdown menu with 'Admin-Group' selected.
- Save** and **Reset** buttons are at the bottom of the form.
- Red numbers 1 through 8 highlight the navigation steps and form fields.

Navigate to **Administration > Identity Management > Identities > Users**. Click **Add**. Provide Name, Passwords and User Groups select Support-Group and click **“Save”** to save the setting.

The screenshot displays the Cisco Identity Services Engine (ISE) Administration console. The navigation path is **Administration > Identity Management > Identities > Users**. The form is titled **New Network Access User**. The fields are as follows:

- Name:** support1
- Status:** Enabled
- Email:**
- Passwords:**
  - \* Login Password:** (masked with dots)
  - Re-Enter Password:** (masked with dots)
  - Enable Password:** (checkbox)
- User Information:**
- Account Options:**
- User Groups:** Support-Group

The **Submit** button is at the bottom.

### Add Network Device:

Navigate to **Administration > Network Resources > Network Devices**. Click **Add**. Provide Name, IP Address, select **TACACS+ Authentication Settings** checkbox and provide Shared Secret key.

The screenshot displays the Cisco Identity Services Engine (ISE) configuration interface for adding a new network device. The navigation path is **Administration > Network Resources > Network Devices**. The left sidebar shows the 'Network devices' list with a 'Default Device'. The main form, titled 'Network Devices List > R1', contains the following fields and options:

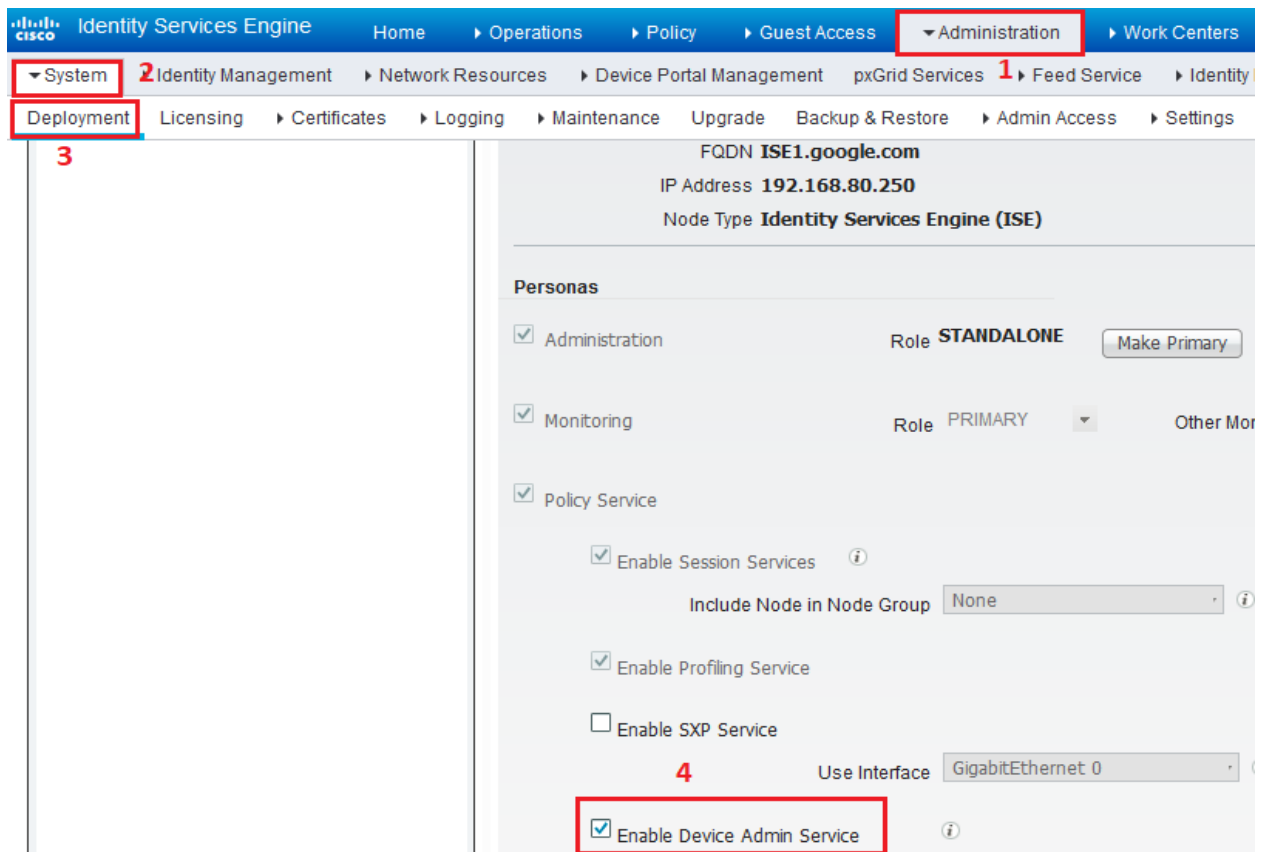
- Name:** R1
- Description:** R1
- \* IP Address:** 192.168.80.201 / 32
- TACACS+ Authentication Settings:** ☒
- Shared Secret:** [Masked with dots] [Show button]
- Enable Single Connect Mode:** ☐
- Device Type:** ☒ Legacy Cisco Device, ☐ TACACS+ Draft Compliance Single Connect Support
- SNMP Settings:** ☐
- Advanced TrustSec Settings:** ☐
- Buttons:** Save, Reset

Navigate to **Administration > Network Resources > Network Devices**. Click **Add**. Provide Name, IP Address, select **TACACS+ Authentication Settings** checkbox and provide Shared Secret key.

The screenshot displays the Cisco Identity Services Engine (ISE) Administration interface. The top navigation bar includes links for Home, Operations, Policy, Guest Access, and Administration (1). The left sidebar shows the navigation menu with System, Identity Management, Network Resources (2), Device Portal Management, pxGrid Services, and Feed S. The Network Resources section is expanded, showing Network Devices (3), Network Device Groups, Network Device Profiles, External RADIUS Servers, and RADIUS Server Seq. The Network Devices list (4) shows a single device named 'S1'. The configuration form for 'S1' (5) includes fields for \* Name (S1), Description (S1), and \* IP Address (192.168.80.202 / 32) (6). The TACACS+ Authentication Settings section (7) is expanded, showing a checked checkbox (8) and a Shared Secret field (9). The Shared Secret field is masked with dots and has a 'Show' button. Below the Shared Secret field, there is a checkbox for 'Enable Single Connect Mode' and two radio buttons for 'Legacy Cisco Device' and 'TACACS+ Draft Compliance Single Connect Support'. At the bottom, there are 'Save' and 'Reset' buttons.

### Enable Device Admin Service:

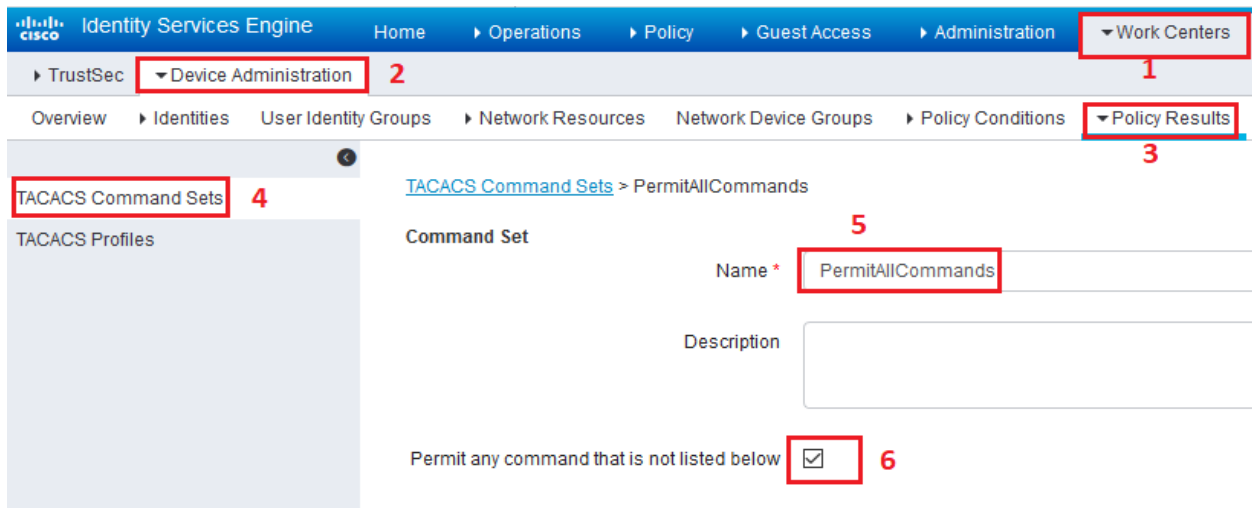
Navigate to **Administration > System > Deployment**. Select required Node. Select **Enable Device Admin Service** checkbox and click **"Save"** to save the setting:



### Configuring TACACS Command Sets:

First **PermitAllCommands** for the user admin, which allow all commands on the device. Second **PermitShowCommands** for user support, which will allow only show commands.

1. Navigate to **Work Centers > Device Administration > Policy Results > TACACS Command Sets**. Click **Add**. Provide the Name **PermitAllCommands**, select **Permit any command** checkbox that is not listed below and click **Submit**.





2. Navigate to **Work Centers > Device Administration > Policy Results > TACACS Command Sets**. Click **Add**. Provide the Name **PermitShowCommands**, click **Add** and permit **show** and **exit** commands. By default if Arguments is left blank, all arguments are included. Click **Submit**.

The screenshot shows the Cisco ISE web interface. The navigation path is: **Work Centers** (1) > **Device Administration** (2) > **Policy Results** (3) > **TACACS Command Sets** (4). The 'PermitShowCommands' command set is being configured (5). The 'Name' field is 'PermitShowCommands'. The 'Description' field is empty. The 'Permit any command that is not listed below' checkbox is unchecked. A table lists commands to be permitted (6):

| Grant                    | Command | Arguments |
|--------------------------|---------|-----------|
| <input type="checkbox"/> | PERMIT  | exit      |
| <input type="checkbox"/> | PERMIT  | show      |

### Configuring TACACS Profile:

Navigate to **Work Centers > Device Administration > Policy Results > TACACS Profiles**. Click **Add**. Provide Name **ShellProfile**, select **Default Privilege** checkbox and enter the value of 15. Click **Submit**.

The screenshot shows the Cisco ISE web interface. The navigation path is: **Work Centers** (1) > **Device Administration** (2) > **Policy Results** (3) > **TACACS Profiles** (4). The 'ShellProfile' TACACS Profile is being configured (5). The 'Name' field is 'ShellProfile'. The 'Description' field is empty. The 'Task Attribute View' tab is selected. The 'Common Tasks' section shows the 'Default Privilege' checkbox checked (6) and the value '15' entered in the adjacent field. A dropdown menu is set to 'Select 0 to 15'.

## Configuring TACACS Authorization Policy:

Navigate to **Work Centers > Device Administration > Policy Sets > Default > Authorization Policy > Edit > Insert New Rule Above**.

Two authorization rules are configured, first rule assigns TACACS profile **ShellProfile** and command Set **PermitAllCommands** based on Network **Admin-Group** membership. Second rule assigns TACACS profile **ShellProfile** and command Set **PermitShowCommands** based on Network **Support-Group** membership.

Define the Policy Sets by configuring rules based on conditions. Drag and drop sets on the left hand side to change the order. For Policy Export go to Administration > System > Backup & Restore > Policy Export Page

| Status | Name    | Description    |
|--------|---------|----------------|
| ✓      | Default | Tacacs_Default |

Regular ☐ Proxy Sequence ☐

Proxy Server Sequence

Authentication Policy

Authorization Policy

Exceptions (0)

| Status | Rule Name          | Conditions (identity groups and other conditions) | Command Sets                             | Shell Profiles |
|--------|--------------------|---------------------------------------------------|------------------------------------------|----------------|
| ✓      | PermitAllCommands  | if Admin-Group                                    | then PermitAllCommands AND ShellProfile  |                |
| ✓      | PermitShowCommands | if Support-Group                                  | then PermitShowCommands AND ShellProfile |                |
| ✓      | Tacacs_Default     | if no matches, then                               | DenyAllCommands                          |                |

## Verify:

Navigate to **Operations > TACACS LiveLog**.

| Logged Time             | Status | Details | Username | Type           | Authentication Policy                | Authorization Policy              | ISE Node |
|-------------------------|--------|---------|----------|----------------|--------------------------------------|-----------------------------------|----------|
| 2019-06-07 08:54:40.406 | ✗      |         | support1 | Authorization  |                                      | Tacacs_Default >> PermitShowC...  | ISE1     |
| 2019-06-07 08:54:38.46  | ✓      |         | support1 | Authorization  |                                      | Tacacs_Default >> PermitShowC...  | ISE1     |
| 2019-06-07 08:54:38.417 | ✓      |         | support1 | Authentication | Tacacs_Default >> Default >> Default |                                   | ISE1     |
| 2019-06-07 08:54:29.839 | ✓      |         | admin1   | Authorization  |                                      | Tacacs_Default >> PermitAllCom... | ISE1     |
| 2019-06-07 08:54:23.9   | ✓      |         | admin1   | Authorization  |                                      | Tacacs_Default >> PermitAllCom... | ISE1     |
| 2019-06-07 08:54:21.138 | ✓      |         | admin1   | Authorization  |                                      | Tacacs_Default >> PermitAllCom... | ISE1     |
| 2019-06-07 08:54:18.395 | ✓      |         | admin1   | Authorization  |                                      | Tacacs_Default >> PermitAllCom... | ISE1     |
| 2019-06-07 08:54:18.296 | ✓      |         | admin1   | Authentication | Tacacs_Default >> Default >> Default |                                   | ISE1     |

## Cisco IOS Router Verification.

```
R1#telnet 192.168.80.202
Trying 192.168.80.202 ... Open
Username:support1
Password:
```

```
SW1#config t
Command authorization failed.
```